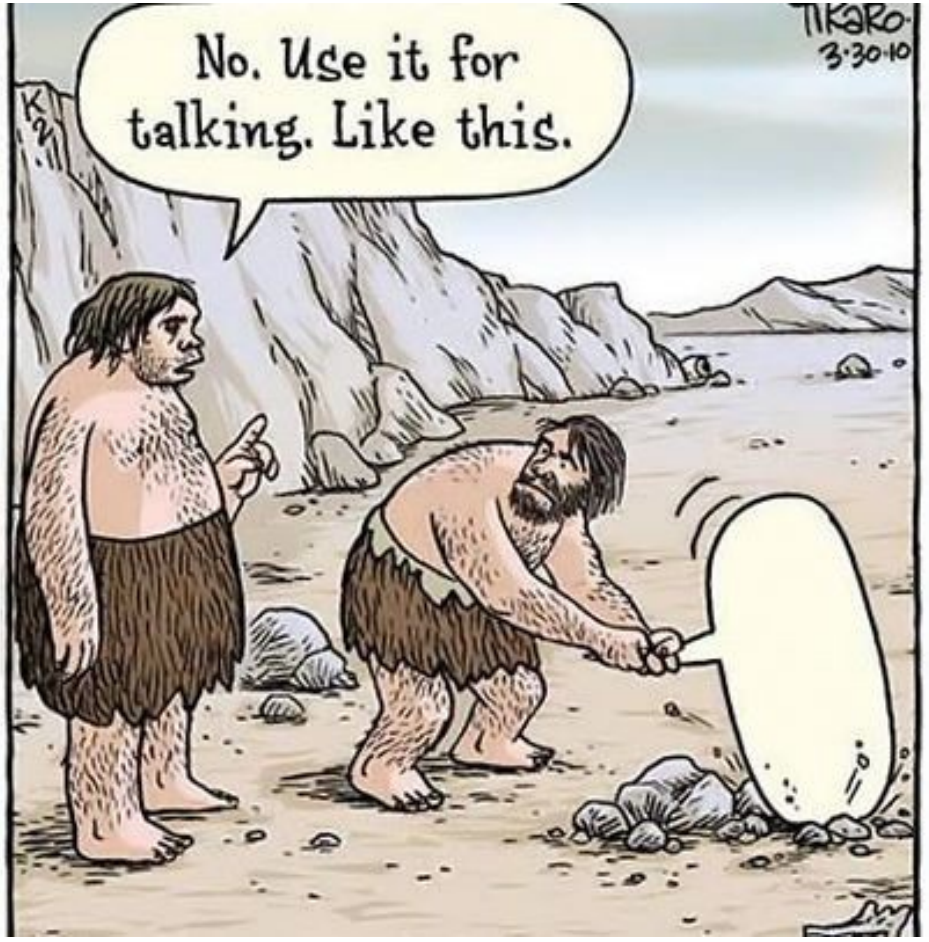
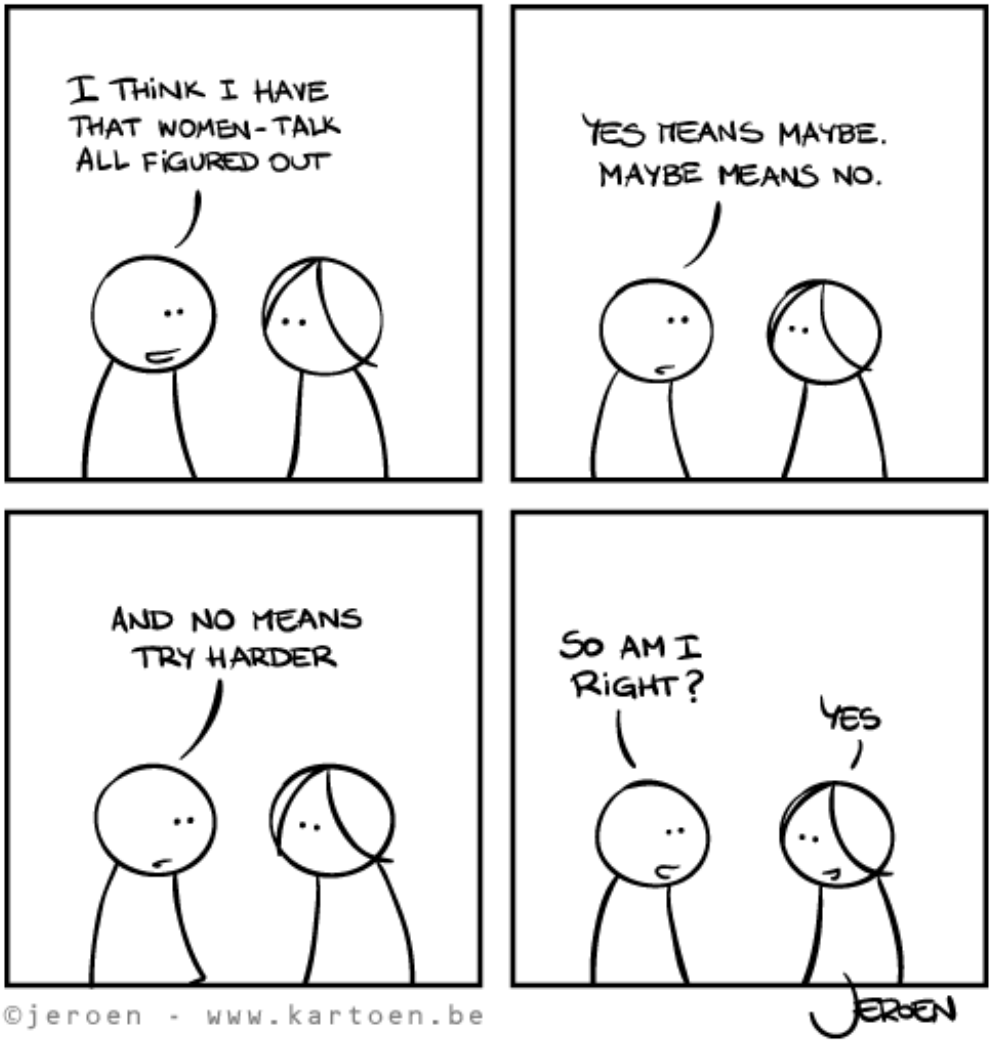
Natural Language Processing

Pragmatics



NLP: Pragmatics



Pragmatics

Pragmatics is a subfield of linguistics and semiotics* that studies the ways in which context contributes to meaning. Pragmatics encompasses speech act theory, conversational implicature, talk in interaction and other approaches to language behavior in philosophy, sociology, linguistics and anthropology.

Pragmatics studies how the transmission of meaning depends not only on structural and linguistic knowledge (e.g., grammar, lexicon, etc.) of the speaker and listener, but also on the context of the utterance, any pre-existing knowledge about those involved, the inferred intent of the speaker, and other factors.

*Semiotics (also called semiotic studies) is the study of meaning-making, the study of sign process (semiosis) and meaningful communication.

Some slides from the S&DAI course

(very quick introduction to “communication as an action”, to limit overlaps between the courses as much as possible....)

Communication

Communication, in general, is the intentional exchange of information brought about by the production and perception of signs drawn from a shared system of conventional signs.

"Classical" view (pre-1953):

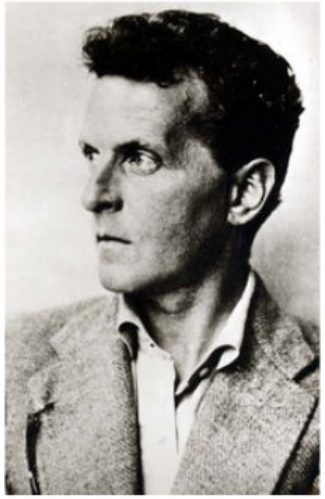
language consists of sentences that are true/false (cf. logic)

"Modern" view (post-1953):

language is a form of action

Communication

Wittgenstein (1953) **Philosophical Investigations**

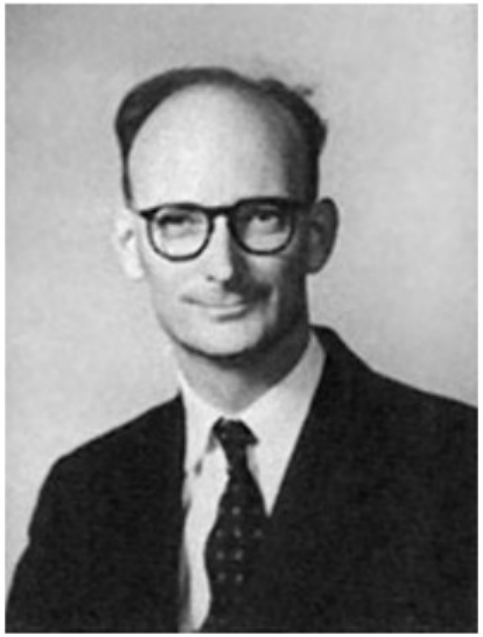


According to Wittgenstein,

- The structure of reality determines the structure of language.
- Meaning is determined by use, in the context of a "language-game".
- Words are not defined by reference to the objects or things which they designate in the external world nor by the thoughts, ideas, or mental representations that one might associate with them, but rather by how they are used in effective, ordinary communication.

Communication

Austin (1962) **How to Do Things with Words**



- In saying something, we do something, such as when a minister joins two people in marriage saying "I now pronounce you husband and wife".

Communication

Searle (1969) **Speech Act Theory**



- Each illocution can be described in terms of what it is attempting to do. So an assertion counts as an undertaking that something really is the case. A question counts as an attempt to elicit some information. Thanking someone counts as an expression of gratitude.
- Assumed intent of the speaker, or the *intentionality* of the sentence.

Dibris

Speech acts



Speech acts achieve the speaker's goals:

Inform	"There's a pit in front of you"
Query	"Can you see the gold?"
Command	"Pick it up"
Promise	"I'll share the gold with you"
Acknowledge	"OK"

Speech act planning requires knowledge of

- Situation
- Semantic and syntactic conventions
- Hearer's goals, knowledge base, and rationality

Dibris

Stages in communication

Intention	S wants to inform H that P
Generation	S selects words W to express P in context C
Synthesis	S utters words W
Perception	H perceives W' in context C'
Analysis	H infers possible meanings $P_1, \ldots P_n$
Disambiguation	H infers intended meaning P_i
Incorporation	H incorporates P_i into KB

Dibris

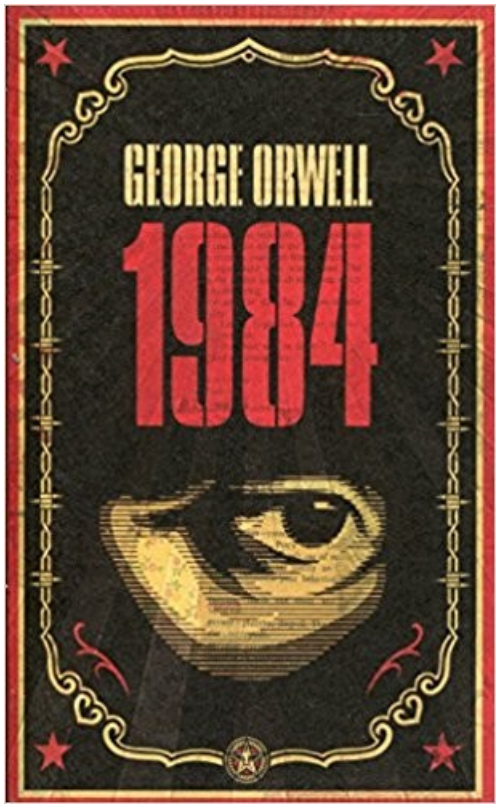
Pragmatics

Thoughts shape language...

but...

...Does Language Shape Thought?

...Does Language Shape Thought?



Nineteen Eighty-Four by George Orwell, June 1949

...Does Language Shape Thought?

"Don't you see that the whole aim of Newspeak is to narrow the range of thought? In the end we shall make thoughtcrime literally impossible, because there will be no words in which to express it. Every concept that can ever be needed will be expressed by exactly one word, with its meaning rigidly defined and all its subsidiary meanings rubbed out and forgotten. Already, in the Eleventh Edition, we're not far from that point. But the process will still be continuing long after you and I are dead. Every year fewer and fewer words, and the range of consciousness always a little smaller. Even now, of course, there's no reason or excuse for committing thoughtcrime. It's merely a question of self-discipline, reality-control. But in the end there won't be any need even for that. The Revolution will be complete when the language is perfect. Newspeak is Ingsoc and Ingsoc is Newspeak," he added with a sort of mystical satisfaction. "Has it ever occurred to you, Winston, that by the year 2050, at the very latest, not a single human being will be alive who could understand such a conversation as we are having now?"

https://genius.com/George-orwell-nineteen-eighty-four-book-1-chapter-5-annotated

...Does Language Shape Thought?

The hypothesis of linguistic relativity holds that the structure of a language affects its speakers' world view or cognition. Also known as the Sapir–Whorf hypothesis, or Whorfianism, the principle is often defined to include two versions: the strong hypothesis and the weak hypothesis:

- The strong version says that language determines thought and that linguistic categories limit and determine cognitive categories.
- The weak version says that linguistic categories and usage only influence thought and decisions.

Edward Sapir (January 26, 1884 - February 4, 1939); Benjamin Lee Whorf (April 24, 1897 - July 26, 1941)

...Does Language Shape Thought?

Linguistic determinism

- The strongest form of the theory is linguistic determinism, which holds that language entirely determines the range of cognitive processes. The hypothesis of linguistic determinism is now generally agreed to be false.

Linguistic influence

- This is the weaker form, proposing that language provides constraints in some areas of cognition, but that it is by no means determinative. Research on weaker forms has produced positive empirical evidence for a relationship.

...Does Language Shape Thought?

Based on the talk by Lera Boroditsky available at https://www.youtube.com/watch?v=iGuuHwbuQOg

Lera Boroditsky is an Associate Professor of Cognitive Science at UCSD and Editor in Chief of Frontiers in Cultural Psychology. She previously served on the faculty at MIT and at Stanford. Her research is on the relationships between mind, world, and language (or how humans get so smart).




How languages & cultures
shape the ways we think
Lera Boroditsky
UC San Diego
SAR

Trump read Ta-Nahisi's latest book

did this happen in the past?
Donald or Melania?
was the action completed?
did you witness this with your own eyes?



the 7th blue house to the left of the big tree

Seventh?
Blue? Light or dark blue? Grue?
Left of? North-Northwest? Volcanoward?
Tree? Banyan? Red cypress? Eucalyptus?



How many languages in the world?

How many languages in the world?

How many languages are there in the world?

7,168 languages are in use today. That number is constantly in flux, because we're learning more about the world's languages every day. And beyond that, the languages *themselves* are in flux. They're living and dynamic, used by communities whose lives are shaped by our rapidly changing world. This is a fragile time: Roughly 40% of languages are now endangered, often with fewer than 1,000 users remaining. Meanwhile, just 23 languages account for more than half the world's population.



https://www.ethnologue.com/guides/how-many-languages (October 24th,2023)

How many languages in the world?

How many languages are there in the world?

7,164 languages are in use today.

That number is constantly in flux, because we're learning more about the world's languages every day. And beyond that, the languages *themselves* are in flux. They're living and dynamic, used by communities whose lives are shaped by our rapidly changing world. This is a fragile time: Roughly 44% of all languages are now endangered, often with fewer than 1,000 users remaining. Meanwhile, the world's 20 largest languages are the native tongue of more than 3.6 billion people total. That's just 0.3% of the world's languages accounting for nearly half of the world's population!



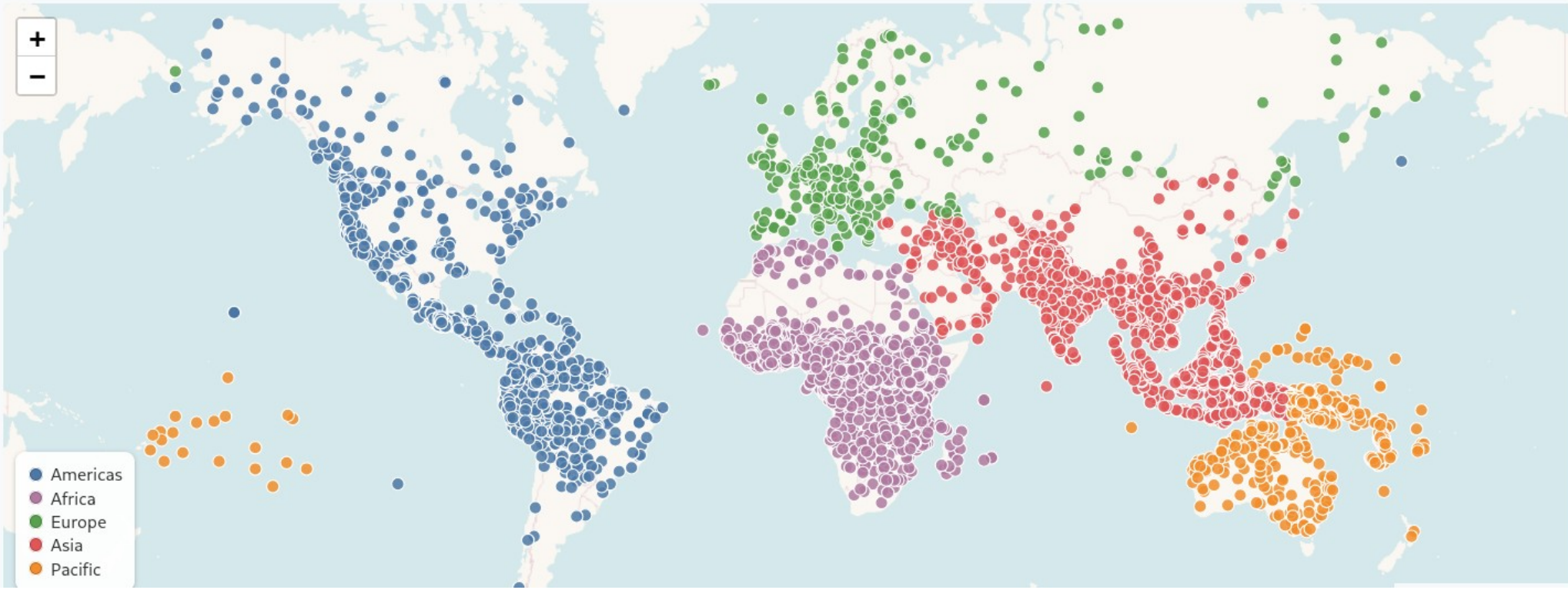
Ethnologue.com
(October 30th,2024)

How many languages in the world?

How Many Languages Are There In The World?

7,159 languages are in use today.

That number is constantly in flux, because we're learning more about the world's languages every day. And beyond that, the languages *themselves* are in flux. They're living and dynamic, used by communities whose lives are shaped by our rapidly changing world. This is a fragile time: Roughly 44% of all languages are now endangered, often with fewer than 1,000 users remaining. Meanwhile, the world's 20 largest languages are the native tongue of more than 3.7 billion people total. That's just 0.3% of the world's languages accounting for nearly half of the world's population!



Ethnologue.com (November 12th,2025)

How many languages in the world?



Other sources: https://wals.info/languoid/genealogy

Navajo verb stems of consumption

	stem	*manner/food of consumption*	*example edibles*
i.	*-yíį́h*	consume/chew UNSPECIFIED	Anything
ii.	*-'aaɬ*	consume/chew HARD, COMPACT	corn, peyote, candy
iii.	*-chozh*	consume/chew LEAFY	lettuce, hay
iv.	*-ghaɬ*	consume/chew MEAT	meat
v.	*-keed*	consume/chew ROUND	bun, melon
vi.	*-ts'ééh*	consume/chew MUSHY MATTER	mush, jello
vii.	*-deeɬ*	consume/chew PLURAL OBJECT	berries, eggs
viii.	*-wol*	consume/chew MARROW	marrow
ix.	*-jooɬ*	consume NON-COMPACT MATTER	cotton candy
x.	*-tłeeh*	consume MUSHY MATTER	ice cream
xi.	*-mááɬ*	consume/devour by GULPING/BOLTING	single piece of food
xii.	*-'aah*	consume SOLID, ROUND OBJECT dunked in liquid	moistened bread, cake
xiii.	*-dlá*	consume/drink UNSPECIFIED	soup, thin gruel
xiv.	*-kaah*	consume/drink from OPEN CONTAINER	milk from glass
xv.	*-t'aah*	consume/drink from CLOSED CONTAINER	milk from bottle



To have a second language is to have a second soul.

- Charlemagne, Holy Roman Emperor

A man who knows four languages is worth four men.

- Charles V, Holy Roman Emperor

I speak English to my accountants, French to my ambassadors, Italian to my mistress, Latin to my God, and German to my horse.

- Frederick the Great of Prussia

I hate relativism more than I hate anything else, excepting, maybe, fiberglass powerboats.

- Jerry Fodor



Jerry Fodor

Jerry Alan Fodor (1935-2017) was an American philosopher and cognitive scientist. He holds the position of State of New Jersey Professor of Philosophy, Emeritus, at Rutgers University and is the author of many works in the fields of philosophy of mind and cognitive science. He is known as "one of the principal philosophers of mind of the late twentieth and early twenty-first century. In addition to having exerted an enormous influence on virtually every portion of the philosophy of mind literature since 1960, Fodor's work has had a significant impact on the development of the cognitive sciences."

Most common words in...

English...
https://en.wikipedia.org/wiki/Most_common_words_in_English

Italian...
https://www.surfacelanguages.com/language/Italian/commonitalianwords.html?Unit=4
http://telelinea.free.fr/italien/1000_parole.html

French...
http://www.vistawide.com/french/top_french_vocabulary.htm

Top 5 nouns in English

1. time
2. year
3. people
4. way
5. day




Nestlē
Tversky et al (1991); Fuhrman & Boroditsky (2010)



Núñez & Sweetser (2006)

last week





next week

transverse



sagittal



vertical



Boroditsky et al (2010), Cognition; Fuhrman et al (2011), Cognitive Science.



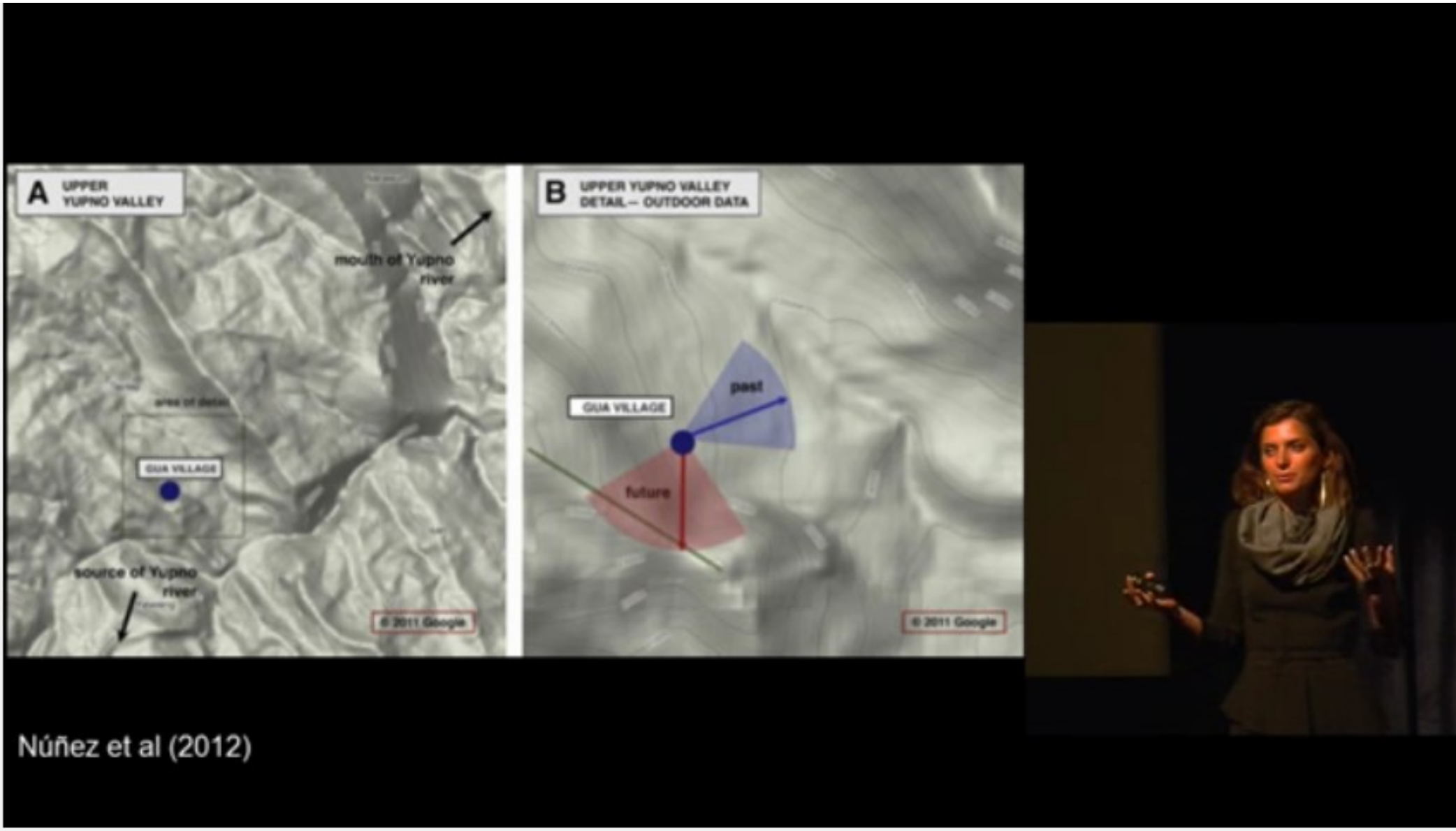




Boroditsky & Gaby (2010). Psychological Science.




Boroditsky & Gaby (2010). Psych Science.
SAR



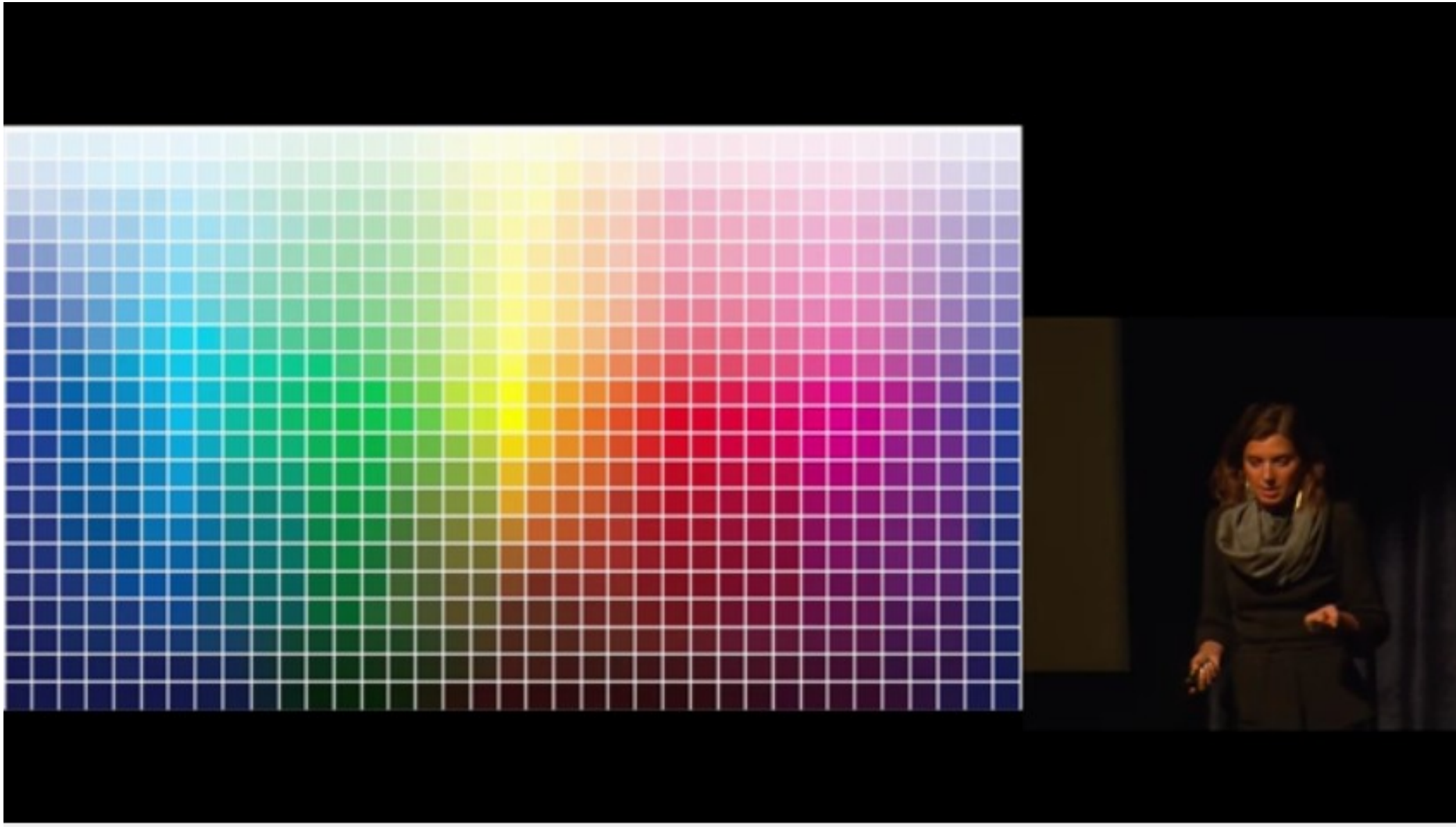
Núñez et al (2012)

How do we know language is causal?

- Teach people new metaphors for time
- See if they change the way they think

- Answer: yes






blue
goluboy
siniy
Winawer, Witthoft, Wu, Frank, Wade, Boroditsky. PNAS.

		
German	Feminine	Masculine
Spanish	Masculine	Feminine



The pervasiveness of gender

- Different phonology
  - okno (-o -e = neuter), taburetka (-a = feminine), stul (consonant = masculine)
- Different number words
  - odna taburetka vs odin stul
- Different adjective endings
  - beloye okno, belaya taburetka, belly stul,
- Different pronouns & possessives
  - ono beloye, ona belaya, on belly
  - moyo okno beloye, moya taburetka belaya, moy stul belly
- Different verb endings
  - moyo okno bilo beloye, moya taburetka bila belaya, moy stul bil belly



Do people really take these genders as meaningful?

- give voices to animated characters
- describe objects
- underestimate hurricanes w/ female names
- personify abstract entities in art





Michelangelo, Parti della Giornata, sarcofago della tomba di Giuliano de' Medici duca di Nemours, 1526-1531

Pat broke the vase.



Pat rompió el florero.



Se rompió el florero.





Who did it the first time?







Language Primes





"The match blew out"
vs
"She blew out the match"



Language Primes





"The toast burned"
vs
"She burned the toast"





Who did it the first time?







A few words on maths

"Philosophy is written in this grand book, which stands continually open before our eyes (I say the 'Universe'), but can not be understood without first learning to comprehend the language and know the characters as it is written. It is written in mathematical language, and its characters are triangles, circles and other geometric figures, without which it is impossible to humanly understand a word; without these one is wandering in a dark labyrinth.”

A few words on maths

"Philosophy is written in this grand book, which stands continually open before our eyes (I say the 'Universe'), but can not be understood without first learning to comprehend the language and know the characters as it is written. It is written in mathematical language, and its characters are triangles, circles and other geometric figures, without which it is impossible to humanly understand a word; without these one is wandering in a dark labyrinth."

Il Saggiatore, Galileo Galilei (1564-1642), fisico, astronomo, scrittore.

A few words on maths

"La filosofia naturale è scritta in questo grandissimo libro che continuamente ci sta aperto innanzi agli occhi, io dico l'universo, ma non si può intendere se prima non s'impara a intender la lingua e conoscer i caratteri nei quali è scritto. Egli è scritto in lingua matematica, e i caratteri son triangoli, cerchi ed altre figure geometriche, senza i quali mezzi è impossibile a intenderne umanamente parola; senza questi è un aggirarsi vanamente per un oscuro labirinto."

Il Saggiatore, Galileo Galilei (1564-1642), fisico, astronomo, scrittore.

A few words on maths

This article is about the Greek scholar of the third century BC. For the crater named after him, see Eratosthenes (crater). For the ancient Athenian statesman of the fifth century BC, see Eratosthenes (statesman).

Eratosthenes of Cyrene (/ɛrəˈtɒsθəniːz/; Greek: Ἐρατοσθένης ὁ Κυρηναῖος, IPA: [eratostʰénɛːs]; c. 276 BC[1] – c. 195/194 BC[2]) was a Greek mathematician, geographer, poet, astronomer, and music theorist. He was a man of learning, becoming the chief librarian at the Library of Alexandria. He invented the discipline of geography, including the terminology used today.[3]

He is best known for being the first person to calculate the circumference of the Earth, which he did by applying a measuring system using stadia, a standard unit of measure during that time period. His calculation was remarkably accurate. He was also the first to calculate the tilt of the Earth's axis (again with remarkable accuracy). Additionally, he may have accurately calculated the distance from the Earth to the Sun and invented the leap day.[4] He created the first map of the world, incorporating parallels and meridians based on the available geographic knowledge of his era.

Eratosthenes was the founder of scientific chronology; he endeavored to revise the dates of the chief literary and political events from the conquest of Troy. Eratosthenes dated The Sack of Troy to 1183 BC. In number theory, he introduced the sieve of Eratosthenes, an efficient method of identifying prime numbers.

Eratosthenes



Born	276 BC Cyrene
Died	194 BC Alexandria
Occupation	Scholar Librarian Poet Inventor
Known for	Sieve of Eratosthenes Foundation of Geography

A few words on maths

The ancient Greeks originally had a number system like the Romans, but in the 4th century BC, they started using a new system closer to Arabic numbers. Instead of counting I, II, III like the Egyptians or the Romans, they had different symbols for 1,2,3 up to 9, just like us. However, they did not use the same symbols to represent numbers greater than 9. They had a new set of symbols for 10, 20, 30, and so on, and yet another set for 100, 200, 300. This has the disadvantage, like so many of the ancient counting systems, that you eventually ran out of symbols!

The ancient Greeks had no symbol for zero (μηδέν), and did not use a digit placeholder for it. They seemed unsure about the status of zero as a number. They asked themselves, "How can nothing be something?"

A few words on maths



A few words on maths

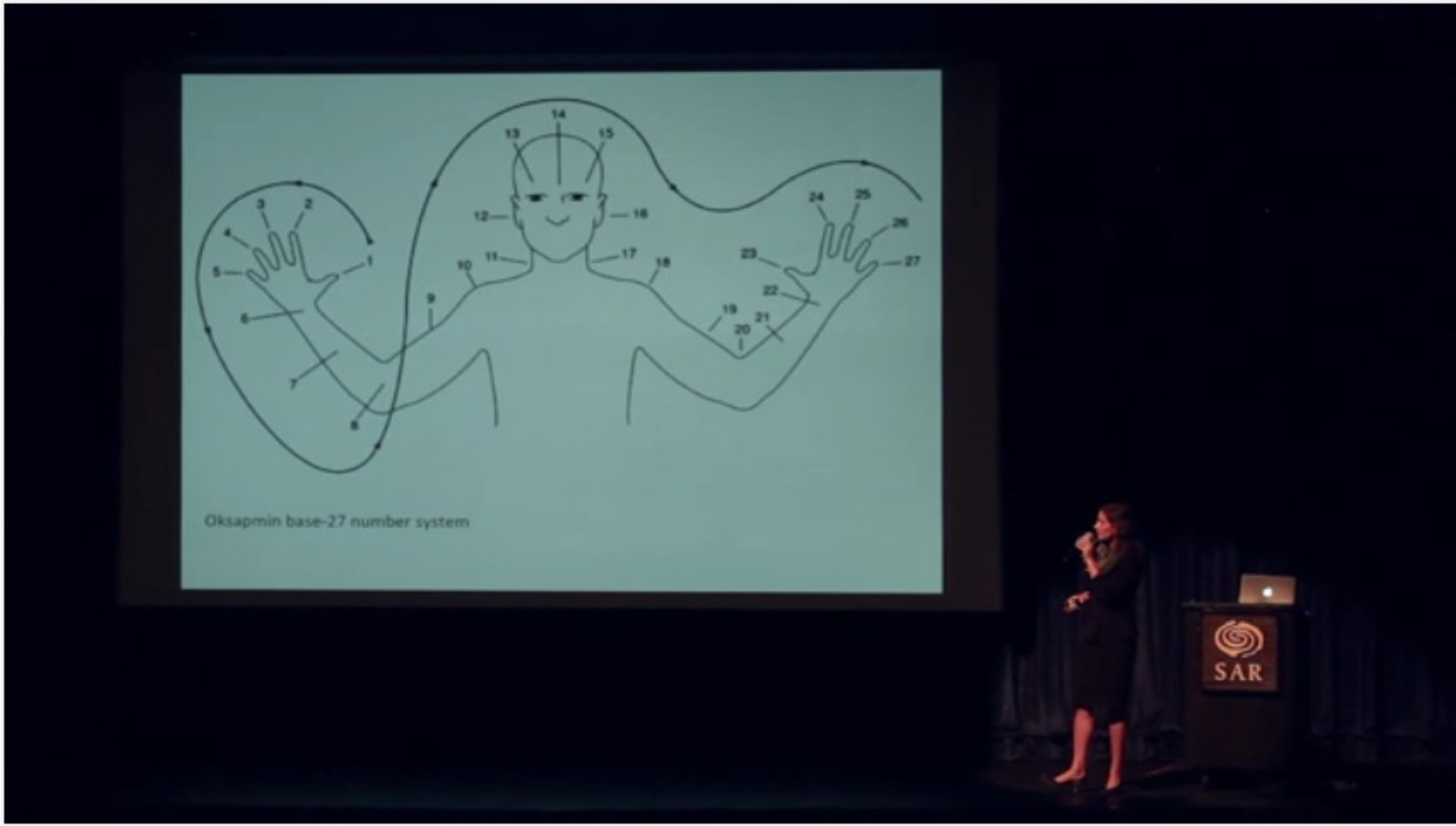
The numeric system represented by Roman numerals originated in ancient Rome and remained the usual way of writing numbers throughout Europe well into the Late Middle Ages. Numbers in this system are represented by combinations of letters from the Latin alphabet. Roman numerals, as used today, are based on seven symbols:

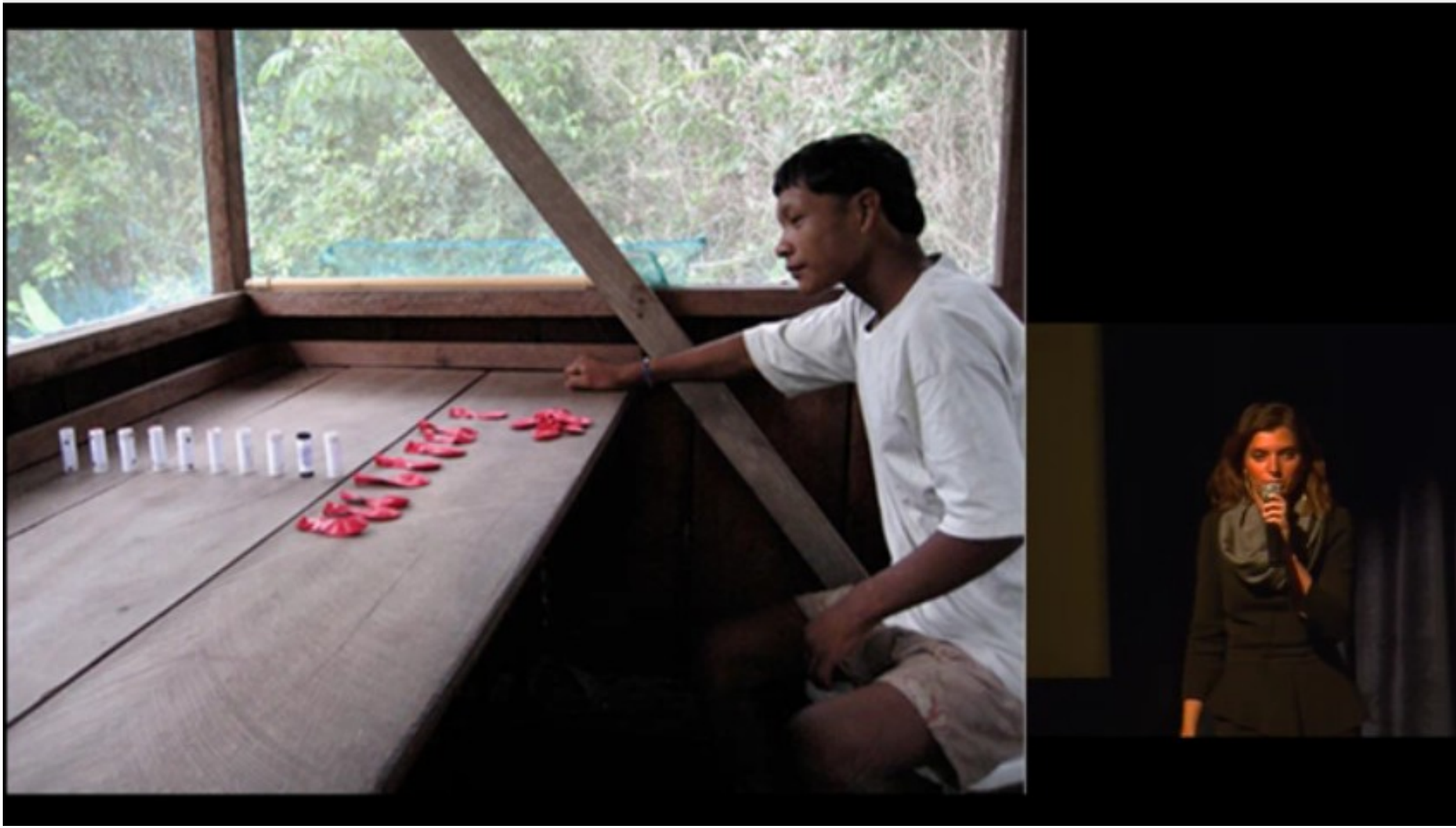
Symbol	I	V	X	L	C	D	M
Value	1	5	10	50	100	500	1,000

The use of Roman numerals continued long after the decline of the Roman Empire. From the 14th century on, Roman numerals began to be replaced in most contexts by the more convenient Hindu-Arabic numerals; however, this process was gradual, and the use of Roman numerals persists in some minor applications to this day.

A few words on maths

Hindu-Arabic numerals, set of 10 symbols -- 1, 2, 3, 4, 5, 6, 7, 8, 9, 0 -- that represent numbers in the decimal number system. They originated in India in the 6th or 7th century and were introduced to Europe through the writings of Middle Eastern mathematicians, especially al-Khwarizmi and al-Kindi, about the 12th century. They represented a profound break with previous methods of counting, such as the abacus, and paved the way for the development of algebra.


Oksapmin base-27 number system
SAR



The importance of what a thing is called

- patriots, activists, or terrorists?
- torture or enhanced interrogation techniques?



But sometimes language seems ineffective. Why?

BBC NEWS WORLD EDITION

Last Updated: Wednesday, 12 March, 2003, 13:40 GMT

Email this to a friend Printable version

US Congress opts for "freedom fries"

French fries in the House of Representatives' cafeterias will now be known as "freedom fries" as part of a Republican protest at France's opposition to a war on Iraq.



French fries are off the menu for US legislators

Republican representative Bob Ney, whose committee is in charge of the eateries, said the action was "a small but symbolic effort to show the strong displeasure of many on Capitol Hill with the actions of our so-called ally, France".

French toast from now on will be known as "freedom toast".



During WWI:

sauerkraut	liberty cabbage
liverwurst	liberty sausage
hamburger	Salisbury steak
Dachshunds	liberty pups

Operation Enduring Freedom:

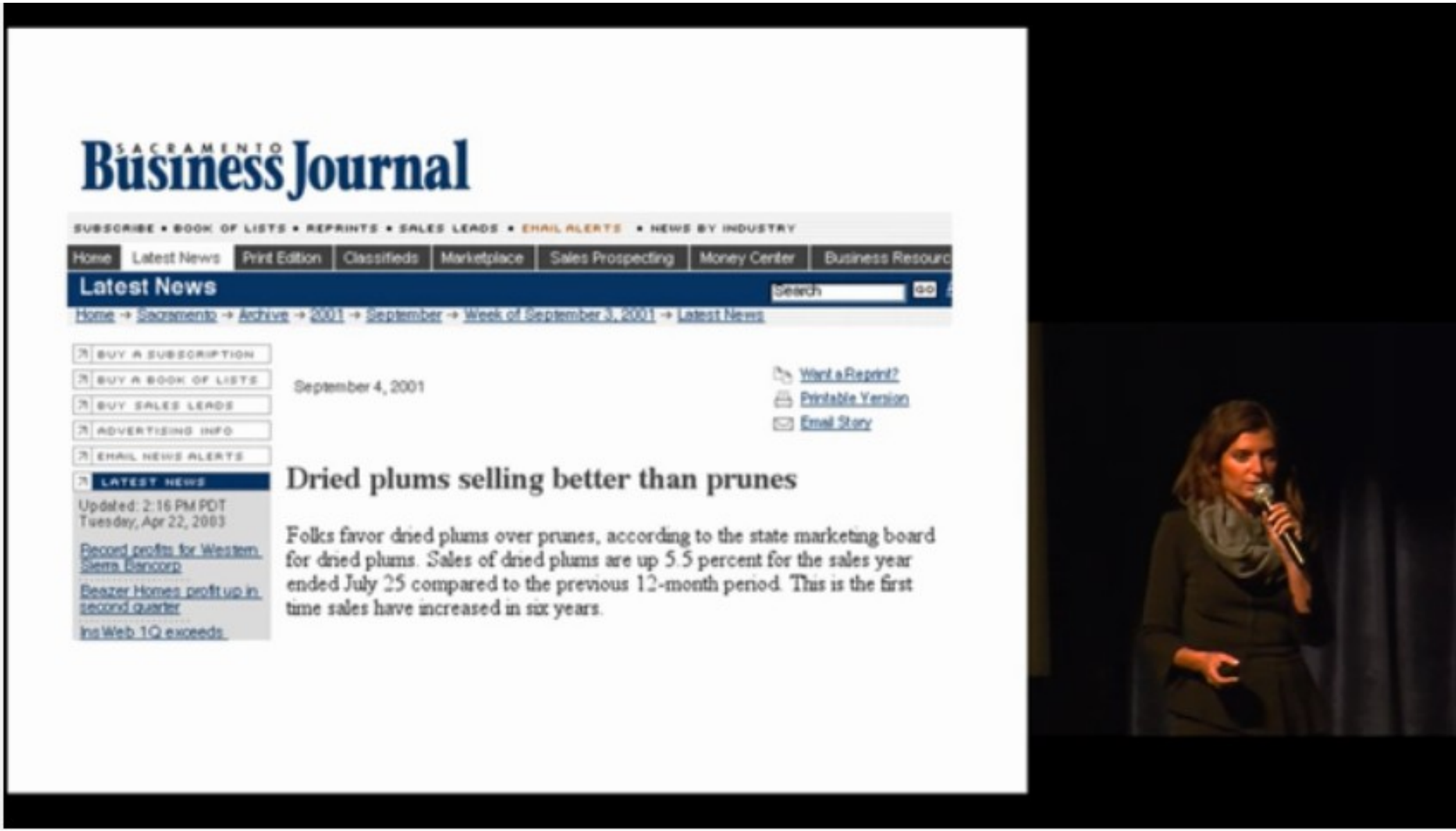
french fries	freedom fries
french toast	freedom toast
french poodles	freedom poodles
french kissing	freedom kissing
french manicures	freedom manicures
France	freedom land?
French	the language of freedom?




SUNSWEET
SELECTED CALIFORNIA
Large Prunes
Good Fiber Source
SEE BACK PANEL FOR NUTRITION INFORMATION
NET WT. 16 OZ. (1 LB.) 454 g


KIRKLAND
SUNSWEET
GOLD LABEL
PITTED DRIED
PLUMS




SACRAMENTO
Business Journal
SUBSCRIBE • BOOK OF LISTS • REPRINTS • SALES LEADS • EMAIL ALERTS • NEWS BY INDUSTRY
Home
Latest News
Print Edition
Classifieds
Marketplace
Sales Prospecting
Money Center
Business Resourc
Latest News
Search
Home → Sacramento → Archive → 2001 → September → Week of September 3, 2001 → Latest News
BUY A SUBSCRIPTION
BUY A BOOK OF LISTS
BUY SALES LEADS
ADVERTISING INFO
EMAIL NEWS ALERTS
LATEST NEWS
Updated: 2:16 PM PDT
Tuesday, Apr 22, 2003
Record profits for Western Sierra Bancorp
Beazer Homes profit up in second quarter
InsWeb 1Q exceeds
September 4, 2001
Want a Reprint?
Printable Version
Email Story
Dried plums selling better than prunes
Folks favor dried plums over prunes, according to the state marketing board for dried plums. Sales of dried plums are up 5.5 percent for the sales year ended July 25 compared to the previous 12-month period. This is the first time sales have increased in six years.

Other examples all around the world
(in Italian)

https://www.focus.it/comportamento/economia/come-si-scelgono-i-nomi-dei-prodotti-e-gli-errori-piu-clamorosi

If you talk to a man in a language he understands, that goes to his head. If you talk to him in his language, that goes to his heart.

Nelson Mandela

